

Privacy-Preserving Natural Language Processing

RUHR
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Lecture 3 – Introduction to Differential Privacy

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CENTER FOR TRUSTWORTHY
DATA SCIENCE AND SECURITY

Intro

- 1 Intro
- 2 Differential privacy basics
- 3 Dataset
- 4 Statistical queries
- 5 Neighboring datasets
- 6 Controlling privacy strength

Overview

First lecture: privacy definitions, arbitrary hard choices

Second lectures: still arbitrary choice in anonymization, but
GDPR notion of possibility of deidentification

Today: towards probabilistic approach to privacy

Differential privacy basics

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Precursors of differential privacy

"Let us begin with a short story. Envision a database of a hospital containing the medical history of some population. On one hand, the hospital would like to advance medical research which is based (among other things) on statistics of the information in the database. On the other hand, the hospital is obliged to keep the privacy of its patients, i.e. leak no medical information that could be related to a specific patient. The hospital needs an access mechanism to the database that allows certain 'statistical' queries to be answered, as long as they do not violate the privacy of any single patient."

I. Dinur and K. Nissim (2003). **"Revealing Information While Preserving Privacy"**. In: *Proceedings of the Twenty-Second ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. SIGMOD/PODS03: International Conference on Management of Data and Symposium on Principles Database and Systems. San Diego California: ACM, pp. 202–210

Precursors of differential privacy

"We focus on binary databases, where the content is of n binary (0-1) entries"

I. Dinur and K. Nissim (2003). **"Revealing Information While Preserving Privacy"**. In: *Proceedings of the Twenty-Second ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. SIGMOD/PODS03: International Conference on Management of Data and Symposium on Principles Database and Systems. San Diego California: ACM, pp. 202–210

First DP papers

"The database consists of some number n of rows. We are completely agnostic about the type of rows – they may be tuples of attributes, strings, or even pictures."

A. Blum, C. Dwork, F. McSherry, and K. Nissim (2005). **"Practical Privacy: The SuLQ Framework"**. In: *Proceedings of the Twenty-Fourth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. SIGMOD/PODS05: International Conference on Management of Data and Symposium on Principles Database and Systems. Baltimore Maryland: ACM, pp. 128–138

Do rows have to be statistically independent?

Dependency between database records

Some research claims that “for providing [the DP] guarantee, differential privacy mechanisms assume independence of tuples in the database.”

F. McSherry (2016). “**Differential Privacy and Correlated Data**”. In: *Personal blog*

But McSherry (2016) refutes this claim:

The [previously quoted assumption about independence] is not correct. No differentially private mechanism assumes anything about the distribution of their input data, precisely because differential privacy does not permit such assumptions.

Dataset

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Dataset (or Database)

The essential assumption (in words)

Each person corresponds to a single row in a table

The semantics of columns is arbitrary:

- It could be an actual "table" with some sensitive attributes (e.g., age, income)
- It could be just an arbitrary unstructured object (e.g., personal photo, text)

Examples of implicit assumptions

Name	Income	Age
Bob	700	21
Alice	2,800	32
Charlie	3,500	45

Table 1: Looks legit, no clear violation of the assumptions

Name	Income	Age	Someone else's income
Bob	700	21	Alice: 2,800
Alice	2,800	32	Charlie: 3,500
Charlie	3,500	45	Alice: 2,800

Table 2: Something looks wrong with this dataset

Examples of implicit assumptions

Name	Income	Age
Bob	700	21
Alice	2,800	32
Charlie	3,500	45
Alice	2,800	32

Table 3: Something looks wrong with this dataset

Statistical queries

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Real-valued query

"Trusted server that holds a database of sensitive information. Given a query function f mapping databases to reals, the so-called true answer is the result of applying f to the database. [...]"

Query

Let X be a database from a universe (a set) of all possible databases $\mathcal{X}$

Real-valued query is

$$f : \mathcal{X} \mapsto \mathbb{R}$$

C. Dwork, F. McSherry, K. Nissim, and A. Smith (2006). **"Calibrating Noise to Sensitivity in Private Data Analysis"**. In: *Theory of Cryptography*. Ed. by S. Halevi and T. Rabin. Red. by D. Hutchison et al. Vol. 3876. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 265–284

How could a counting query leak private information?

"It's just a statistics (a sum), so I can hide in the crowd!"

What is private information

The secret "bit" $f(d_i) \mapsto \{0, 1\}$

which eventually corresponds to whether or not you are in the database

- If your are in the database, you cannot control others
- What if the database is very small?
- What if the database is queried repeatedly?
- What if you are an outlier?

Statistical queries

How to protect privacy in counting queries

Example

Name	Hospitalized in year	Age	Illegal drug use
Alice	2024	32	yes
Bob	2020	21	no
Charlie	2023	45	no
...			
Xander	2020	31	yes

(Important to clarify for context: If you are Alice, how did you end up in such a database?)

Table 4: A sensitive database example from a clinic

Query: How many persons in the database take illegal drugs?

Example: How many persons in the database take illegal drugs?

Name	Hospitalized in year	Age	Illegal drug use
Alice	2024	32	yes
Bob	2020	21	no
Charlie	2023	45	no
...			
Xander	2020	31	yes

What might be public information: the size of the dataset
(say $n = 100$)

The true answer to the query: 2

If Xander reveals his drug use, and we know Alice is in the database, her privacy is revealed!

Statistical queries

How to protect privacy

Privatizing the query result

How would you go about it?

Solution: Alter the query result

Changes to the true query result must be **irreversible!**

"A natural approach, and one that has been explored by others in the 1980's, is to add random noise to the answer" (Blum, Dwork, McSherry, and Nissim, 2005)

"It is evident that without randomness there is no privacy: if everything is pre-determined, and all possible choices we make are predictable or pre-programmed by our adversaries, then there is nothing that we can build our privacy on." (Ekert and Renner, 2014)

A. Blum, C. Dwork, F. McSherry, and K. Nissim (2005). **"Practical Privacy: The SuLQ Framework"**. In: *Proceedings of the Twenty-Fourth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*. SIGMOD/PODS05: International Conference on Management of Data and Symposium on Principles Database and Systems. Baltimore Maryland: ACM, pp. 128–138

A. Ekert and R. Renner (Mar. 2014). **"The Ultimate Physical Limits of Privacy"**. In: *Nature* 507.7493, pp. 443–447

How to randomize the query result?

Recall: The counting query output is a natural number

How should we randomize it?

- Draw a random value from a probability distribution
- Which one? How parametrized?

What we want

The randomized output will be likely close to the true value,
and unlikely far away

The Laplace distribution

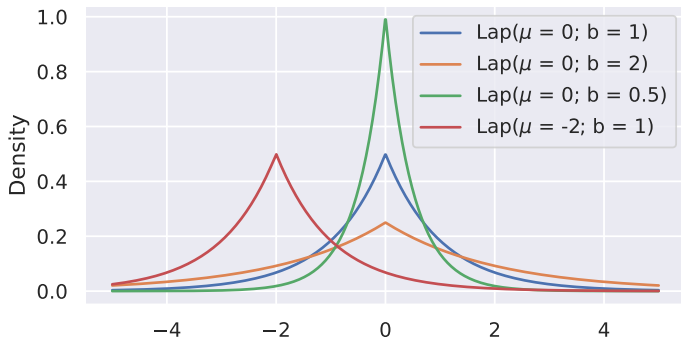


Figure 1: Laplace PDF (probability density function)

$$\text{Lap}(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right)$$

How to choose the parameters?

Location and scale

- Location: the true value
- Scale?

Let's pick $b = 1$ just for lack of better ideas :)

It's all nice and random, but what does it give us?

Query: How many persons in the database take illegal drugs?

Name	Hospitalized in year	Age	Illegal drug use
Alice	2024	32	yes
Bob	2020	21	no
Charlie	2023	45	no
...			
Xander	2020	31	yes

Table 5: Database D , including Alice

Assume that the true answer is 51

It's all nice and random, but what does it give us?

What if Alice decided **not to be part of the data**?

– She has the right to decide about her privacy, GDPR, etc.

Name	Hospitalized in year	Age	Illegal drug use
Bob	2020	21	no
Charlie	2023	45	no
...			
Xander	2020	31	yes

Table 6: Database D' , **excluding** Alice

The true answer to the query would be now 50

We have two databases: With and without Alice

Privatize for D (with Alice), true answer is 51

$$Y \sim \frac{1}{2b} \exp\left(\frac{-|x - 51|}{b}\right) = \frac{1}{2} \exp(-|x - 51|)$$

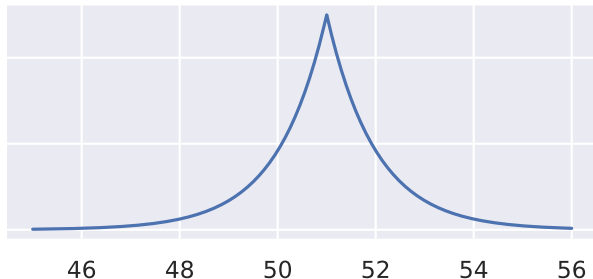


Figure 2: Laplace density, $\mu = 51$, $b = 1$

Now without Alice

Privatize for D' (without Alice), true answer is 50

$$Y' \sim \frac{1}{2b} \exp\left(\frac{-|x - 50|}{b}\right) = \frac{1}{2} \exp(-|x - 50|)$$

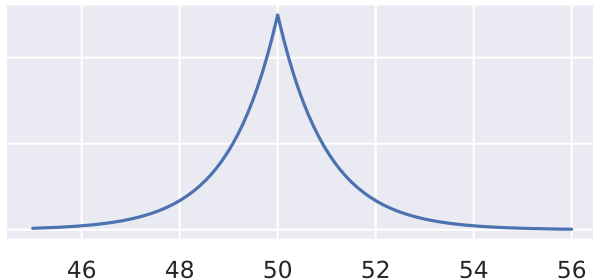


Figure 3: Laplace density, $\mu = 50$, $b = 1$

Both variables in one plot

$$Y \sim \frac{1}{2} \exp(-|x - 51|) \quad Y' \sim \frac{1}{2} \exp(-|x - 50|)$$

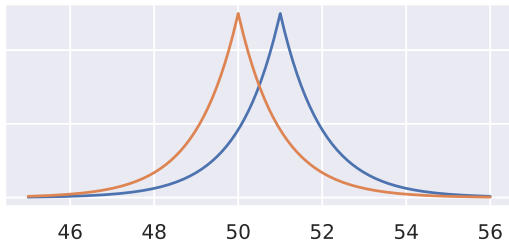


Figure 4: Blue = with Alice, red = without Alice

Now observe a particular value $y \in \mathbb{R}$, for example 52

– Was it sampled from Y or from Y' ?

If we observed 52, was it sampled from Y or from Y' ?

We cannot really tell!

But we can compare probabilities of this event¹

$$\Pr[Y = 52] = \frac{1}{2} \exp(-|52 - 51|) = 0.5 \exp(-1)$$

$$\Pr[Y' = 52] = \frac{1}{2} \exp(-|52 - 50|) = 0.5 \exp(-2)$$

How much more likely from Y ?

$$\frac{\Pr[Y=52]}{\Pr[Y'=52]} = \frac{0.5 \exp(-1)}{0.5 \exp(-2)} = \exp(-1) \exp(2) = \exp(1) = 2.718$$

How much more likely from Y' ?

$$\frac{\Pr[Y'=52]}{\Pr[Y=52]} = \frac{0.5 \exp(-2)}{0.5 \exp(-1)} = \exp(-1) = 0.367$$

¹We must compare densities, so our math will not break

Both variables in one plot

$$Y \sim \frac{1}{2} \exp(-|x - 51|) \quad Y' \sim \frac{1}{2} \exp(-|x - 50|)$$

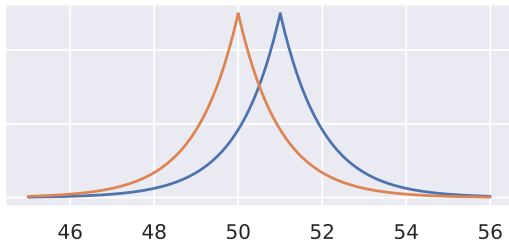


Figure 5: Blue = with Alice, red = without Alice

Now observe a particular value $y \in \mathbb{R}$, for example 50

– Was it sampled from Y or from Y' ?

If we observed 50, was it sampled from Y or from Y' ?

$$\Pr[Y = 50] = \frac{1}{2} \exp(-|50 - 51|) = 0.5 \exp(-1)$$

$$\Pr[Y' = 50] = \frac{1}{2} \exp(-|50 - 50|) = 0.5 \exp(0)$$

How much more likely from Y ?

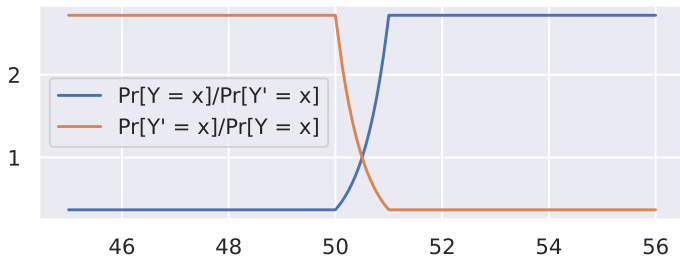
$$\frac{\Pr[Y=50]}{\Pr[Y'=50]} = \frac{0.5 \exp(-1)}{0.5 \exp(0)} = \exp(-1) = 0.367$$

How much more likely from Y' ?

$$\frac{\Pr[Y'=50]}{\Pr[Y=50]} = \frac{0.5 \exp(0)}{0.5 \exp(-1)} = \exp(1) = 2.718$$

It seems like the odds ratio is bounded from above...

Can we generalize it for any observed x ?



Seems like the maximum we can get is $2.718 = e = \exp(1)$

Let's try to prove it

We need triangle inequality for absolute values:²

$$|x| - |y| \leq |x - y|$$

$$\begin{aligned} \frac{\Pr[Y = x]}{\Pr[Y' = x]} &= \frac{\frac{1}{2} \exp(-|x - 51|)}{\frac{1}{2} \exp(-|x - 50|)} = \exp(|x - 50| - |x - 51|) \\ &\leq \exp(|(x - 50) - (x - 51)|) \\ &= \exp(|x - 50 - x + 51|) \\ &= \exp(|1|) = \exp(1) \end{aligned}$$

²Homework: Prove it! $x, y \in \mathbb{R}$

Let's try to prove it (part 2)

$$\begin{aligned}\frac{\Pr[Y' = x]}{\Pr[Y = x]} &= \frac{\frac{1}{2} \exp(-|x - 50|)}{\frac{1}{2} \exp(-|x - 51|)} = \exp(|x - 51| - |x - 50|) \\ &\leq \exp(|(x - 51) - (x - 50)|) \\ &= \exp(|x - 51 - x + 50|) \\ &= \exp(|-1|) = \exp(1)\end{aligned}$$

What does that mean?

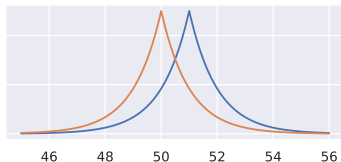


Figure 6: Blue = with Alice, red = without Alice,
 $Y \sim \frac{1}{2} \exp(-|x - 51|)$ $Y' \sim \frac{1}{2} \exp(-|x - 50|)$

$$\frac{\Pr[Y=x]}{\Pr[Y'=x]} \leq \exp(1) \quad \frac{\Pr[Y'=x]}{\Pr[Y=x]} \leq \exp(1)$$

No matter what value we get after privatizing the counting query – we can only get some limited "information" about whether it came from Y or Y' .

What if the true answer would be the same for D and D' ?

Alice did not take drugs

$$Y \sim \frac{1}{2} \exp(-|x - 50|) \quad Y' \sim \frac{1}{2} \exp(-|x - 50|)$$

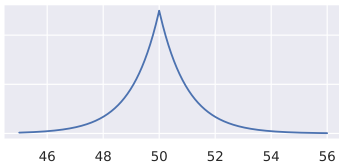


Figure 7: Both have the same distribution

$$\frac{\Pr[Y=x]}{\Pr[Y'=x]} = 1 \leq \exp(1) \quad \frac{\Pr[Y'=x]}{\Pr[Y=x]} = 1 \leq \exp(1)$$

So the maximum "loss" only when the presence of Alice changes the query output

What if the true answer would be 22?

Alice did not take drugs

$$Y \sim \frac{1}{2} \exp(-|x - 22|) \quad Y' \sim \frac{1}{2} \exp(-|x - 22|)$$

– bounded by 1 as in the previous slide

Alice did take drugs

$$Y \sim \frac{1}{2} \exp(-|x - 23|) \quad Y' \sim \frac{1}{2} \exp(-|x - 22|)$$

– our general proof still holds!

$$\frac{\Pr[Y=x]}{\Pr[Y'=x]} \leq \exp(1) \quad \frac{\Pr[Y'=x]}{\Pr[Y=x]} \leq \exp(1)$$

Summary

Four counting queries, the maximum difference of the query result is 1 (when a particular person is not in the dataset)

We used scale $b = 1$ for the Laplace distribution, which gives us upper bound on privacy loss

$$\frac{\Pr[Y=x]}{\Pr[Y'=x]} \leq \exp(1) \quad \frac{\Pr[Y'=x]}{\Pr[Y=x]} \leq \exp(1)$$

Neighboring datasets

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Defining neighboring datasets

We saw that the bound was 'symmetric'

$$\frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \exp(1) \quad \frac{\Pr[Y' = y]}{\Pr[Y = y]} \leq \exp(1)$$

On the left: One entry less in the denominator (= one entry more in the nominator)

On the right: One entry less in the nominator (= one entry more in the denominator)

The bound holds for any two datasets that **differ in the presence of one entry** (e.g., one row removed, or one row added) → **neighboring datasets**

Neighboring dataset examples (choice of Alice is arbitrary!)

Name	Hospitalized in year	Age	Illegal drug use
Alice	2024	32	yes
Bob	2020	21	no
Charlie	2023	45	no

Table 7: Database 1, including Alice

Name	Hospitalized in year	Age	Illegal drug use
Bob	2020	21	no
Charlie	2023	45	no

Table 8: Database 2, excluding Alice

Neighboring dataset examples

Databases 1 and 2 from the previous slide are **neighboring** as they differ in presence or absence of a single row

We denote them 1: D and 2: D' (**an arbitrary choice**)

The neighboring dataset definition allows us to simplify

$$\frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \exp(1) \quad \frac{\Pr[Y' = y]}{\Pr[Y = y]} \leq \exp(1)$$

into one inequality (where again Y is a random variable based on D and Y' is a random variable based on D')

$$\frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \exp(1)$$

Restating our bound with neighboring datasets

Neighboring datasets = presence or absence of a single individual

Our bound $\frac{\Pr[Y=y]}{\Pr[Y'=y]} \leq \exp(1)$ holds for any neighboring datasets (we proved it, but convince yourself again)

So the absence or presence of **any single individual**

- will influence the result of the counting query (but we want to keep this influence small → **utility**)
- for any 'all-mighty' adversary, the likelihood to find out the extra person's 'private bit' (e.g., drug use) is upper bounded (the person wants to keep this bound small →

privacy)

Controlling privacy strength

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What if 2.718 is not strong enough?

For $y \sim y_{\text{true}} + \text{Lap}(b = 1)$

The likelihood for preferring one hypothesis over the other

$$\frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \exp(1) \approx 2.718$$

What would be the minimum of $\frac{\Pr[Y=y]}{\Pr[Y'=y]}$?

What would be the maximum of $\frac{\Pr[Y=y]}{\Pr[Y'=y]}$?

$$1 \leq \frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \infty$$

Why?

Playing with the scale parameter b

For $y \sim y_{\text{true}} + \text{Lap}(\mu = 0; b = 1)$

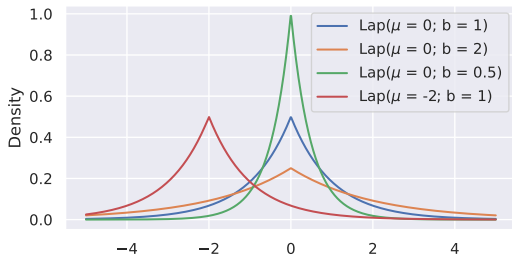


Figure 8: Laplace PDF $\text{Lap}(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$

What to do with the scale for stronger privacy?

Intuition: Larger b gives stronger privacy

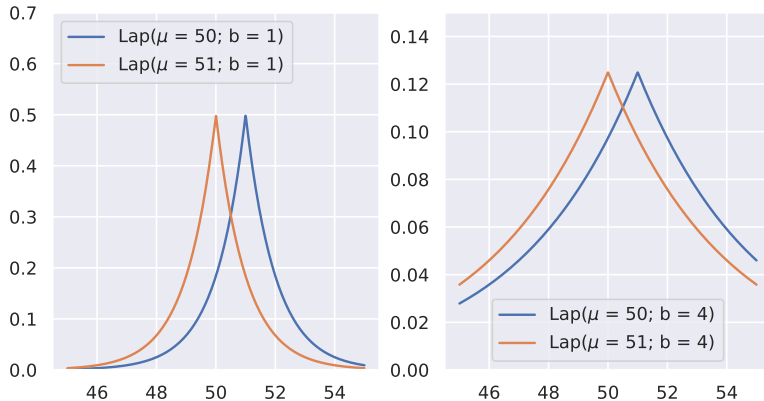


Figure 9: Laplace PDF (probability density function)

How does that relate to the maximum privacy loss?

Recall: likelihood of any output (x -axis) coming from D' as opposed to D (and vice versa)

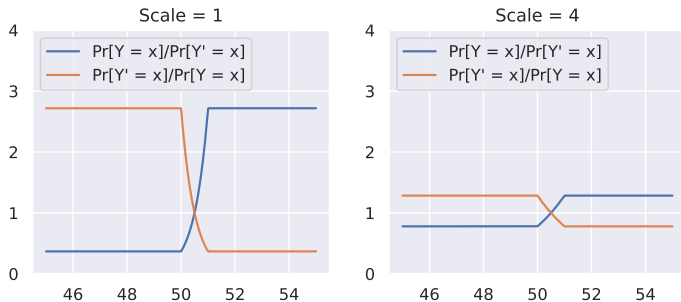


Figure 10: Privacy loss for two Laplace distributions for a counting query, varying scale b

Let's try to formalize our intuition

Counting query true value $y = 51$, one person removed

$y = 50$

$$\begin{aligned}\frac{\Pr[Y = y]}{\Pr[Y' = y]} &= \frac{\frac{1}{2b} \exp\left(-\frac{1}{b}|y - 51|\right)}{\frac{1}{2b} \exp\left(-\frac{1}{b}|y - 50|\right)} = \exp\left(\frac{1}{b}(|y - 50| - |y - 51|)\right) \\ &\leq \exp\left(\frac{1}{b}((y - 50) - (y - 51))\right) \\ &= \exp\left(\frac{1}{b}|y - 50 - y + 51|\right) \\ &= \exp\left(\frac{1}{b}|1|\right) = \exp\left(\frac{1}{b}\right)\end{aligned}$$

Will be the same for $\frac{\Pr[Y'=y]}{\Pr[Y=y]}$ (prove it! check the appendix)

What have we discovered so far

Our choice of 50 and 51 was arbitrary, the same proof will hold for any true answers differing by 1

For a **counting query**, the database curator (**trusted curator**, trusted holder) protects privacy of each individual by reporting

$$y \sim y_{\text{true}} + \text{Lap}(b)$$

This ensures that for **any two neighboring datasets** the privacy loss is **bounded**

$$\frac{\Pr[Y = y]}{\Pr[Y' = y]} \leq \exp\left(\frac{1}{b}\right)$$

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Credits

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